APPENDIX

A. Tracking Policy and Training

This appendix specifies the problem formulation, network architecture, reward function, domain randomization, and training data of the tracking policy described in Secs. IV-A and IV-B.

1) Problem Formulation: We formulate whole-body teleoperation as motion tracking and model it as a Markov decision process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, r)$. The state space $\mathcal{S}$ contains proprioceptive and task-related variables, the action space $\mathcal{A} \subset \mathbb{R}^{29}$ contains target angles for the 29 body joints, $\mathcal{T}$ is the transition dynamics, and the reward r encourages tracking fidelity and physical plausibility. Given a reference trajectory, a policy $\pi : \mathcal{O} \to \mathcal{A}$ maps observations to actions so as to minimize the deviation of the robot from the reference while keeping it balanced. The observation space $\mathcal{O}$ is that of the teacher (Eq. (1)) or of the student (Eq. (2)). We train the teacher with Proximal Policy Optimization (PPO) to maximize the expected cumulative reward and distill the student from it with DAgger.

2) Network Architecture: We use Transformer-based actor-critic networks to model interactions among heterogeneous proprioceptive states and reference commands and, in the student, to draw temporal context from the observation history. The teacher is a two-layer Transformer with embedding dimension $d_{\text{model}} = 256$, four attention heads, and a feed-forward dimension of 512. The student uses $d_{\text{model}} = 512$, four attention heads, and a feed-forward dimension of 1024. Actor and critic heads, each a multi-layer perceptron (MLP) with one hidden layer of 128 units, follow the Transformer features.

The student omits the body-link states $\mathbf{p}_t$ of Eq. (1), which are difficult to estimate reliably on hardware, and must infer from its proprioceptive history (Eq. (2)) what the privileged teacher observes directly. We make the student larger than the teacher because it benefits empirically from extra capacity for this inference.

3) Reward Function and Domain Randomization: Tables V and VI list the reward terms with their weights and the domain-randomization ranges used in training.

4) Training Data: Part of our training data comes from the TWIST2 [16] corpus, whose retargeted motions we expand from 23 to 29 degrees of freedom (DoFs) to match our robot. To align the data with the ***OmniBench*** diagnostics, we filter the reference motions heuristically by root state (position and velocity) and whole-body joint energy. Following the TWIST2 data collection protocol, we then add reference motions captured with our own teleoperation interface. We then retarget these motions with GMR [14]. The final dataset contains 30 hours of motion data, with the same set of in-place manipulation motions shared across all data recipes accounting for approximately 60% of each recipe.

Table VII reports the composition of the three recipe variants and the final recipe over five aggregate motion categories, excluding the shared in-place manipulation motions. Sec. V describes the differences among the recipes. All variants use approximately the same amount of training data, with differences of at most 20 minutes, and the same number of optimization steps, differing only in motion composition.

The variants share a similar overall distribution, in which in-place manipulation and loco-manipulation make up about 78% to 80% of the training data. They therefore differ mainly in the remaining motions, especially in the balance between locomotion and dynamic behaviors. Despite its name, ***OmniClone***^{Dynamic} is not trained only on dynamic motions but has a slightly larger share of highly dynamic and varying-height motions in the shared mixture. Section V-B uses this controlled variation to study how small changes in motion composition affect the learned whole-body behavior.

B. Simulation Experiments

We detail how ***OmniBench*** is generated and scored, compare the two policy backbones group by group, and test how retargeting scale errors affect tracking.

1) Benchmark Generation: The six skills of ***OmniBench*** (Sec. III-A) cover the main operating regimes of teleoperated humanoids, and their three levels of workspace height or dynamic intensity (Tab. I) form a grid over skill type and difficulty for fine-grained diagnosis. We synthesize the held-out reference motions with HY-Motion-1.0 [15] and retarget them directly to the robot with General Motion Retargeting (GMR) [14], without filtering, smoothing, manual correction, or other post-processing. Each skill contains 60 episodes (20 per level) of 90 frames on average, sampled at 30 Hz.

2) Metrics and Baselines: We report success rate (SR) and mean per-joint position error (MPJPE). An episode fails when the pelvis of the robot contacts the ground, and SR is the percentage of episodes that do not fail. MPJPE is the global position error averaged over all body joints and all frames of the episodes, and failed episodes are not included in its computation. All methods are evaluated in MuJoCo under identical simulation settings. The baselines use their officially released code and pretrained checkpoints, and ***OmniClone*** is trained in Isaac Lab (Sec. IV-B).

3) Backbone Ablation: Table VIII isolates the two backbone rows of Tab. II for the ablation of Sec. V-B and marks the better value in each group. The Transformer has lower MPJPE in all 18 groups and equal or higher SR in 17 of them, the exception being medium squatting (95% against 100%). The SR gap is largest in fast and medium walking, where ***OmniClone***^{MLP} succeeds in only 20% and 35% of episodes. We attribute the gain to the ability of the Transformer to model interactions among heterogeneous observation terms and to draw temporal context from the proprioceptive history and the future reference commands.

The backbone is not the only factor, however. The four recipes in Tab. II share the Transformer backbone, yet ***OmniClone***^{Dynamic} reaches a mean SR of only 59.7%, against 97.8% to 98.1% for the other three.

4) Retargeting Scale Discrepancies: Table IX reports the simulation evaluation of scale mismatch summarized in Sec. IV-C. We uniformly scale the motion capture (MoCap) data and retarget the resulting motions using the original, unscaled retargeting configurations, producing scale-induced

TABLE V: **Reward terms and weights.** End-effectors are the wrist-yaw and ankle-roll links. Pos., Rot., Lin., Ang., and Vel. abbreviate position, rotation, linear, angular, and velocity.

Category	Reward Term	Weight
Regularization	Action Rate Penalty	-8.0
	Contact Air Time Penalty	-100.0
	Joint Acceleration Penalty	-1.0×10^{-7}
	Joint Position Limits	-10.0
	Velocity / Action Limits	-1.0
Tracking	Torso Global Pos. / Rot.	0.5
	Full-Body Global Lin. / Ang. Vel.	1.0
	Full-Body Relative Pos. / Rot.	1.0
	End-Effector Relative Pos. / Rot. / Lin. / Ang. Vel.	0.5

TABLE VI: **Primary domain-randomization parameters and ranges.** $\mathcal{U}(a, b)$ is a uniform distribution, nominal values come from the URDF, and CoM and RFI denote center of mass and random force injection. A dash marks a parameter that is not assigned to specific links.

Parameter	Sampling Distribution	Applied Links
Action Delay (s)	$\mathcal{U}(0.0, 0.02)$	–
Action Noise (rad)	$\mathcal{U}(0.0, 0.02)$	–
Link Mass	$[0.9, 1.1] \times$ nominal	Torso, shoulder yaw
Torso CoM Offset (m)	$\delta_x \sim \mathcal{U}(-0.075, 0.075)$, $\delta_{y,z} \sim \mathcal{U}(-0.1, 0.1)$	–
Torque RFI (N·m)	$0.02 \times$ limit	–
Static Friction	$\mathcal{U}(0.3, 2.0)$	Ankle roll, pelvis, hip roll, knee, elbow
Dynamic Friction	$\mathcal{U}(0.3, 2.0)$	Ankle roll, pelvis, hip roll, knee, elbow
Stiffness Scale	$[0.95, 1.05] \times$ nominal	–
Damping Scale	$[0.95, 1.05] \times$ nominal	–
Armature Range	$[0.995, 1.015] \times$ nominal	–

TABLE VII: **Training data composition of the four recipes.** Each entry is the share of a motion category in the full training set of a recipe. ***OmniClone*** without a superscript denotes the final recipe.

Method	High Dynamic	Loco-Manipulation	Locomotion	Varying Height	Low Dynamic
OmniClone^{Balance}	3.96%	42.93%	38.86%	11.09%	3.16%
OmniClone^{Stable}	3.30%	45.16%	39.59%	10.11%	1.84%
OmniClone^{Dynamic}	4.05%	44.30%	38.12%	11.73%	1.80%
OmniClone	3.93%	43.14%	38.90%	11.15%	2.88%

TABLE VIII: **Backbone ablation on *OmniBench*.** SR (%) and MPJPE (mm) of the parameter-matched MLP and Transformer backbones, both trained with the final recipe under identical settings. Manip denotes stationary manipulation. Bold marks the better value in each column, and ties are left unmarked.

(a) Skills stratified by workspace height

Method	**Loco-Manip**						**Manip**						**Squat**					
	High		Medium		Low		High		Medium		Low		High		Medium		Low	
	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓
OmniClone^{MLP}	85	58.9	65	75.4	65	73.4	100	36.3	100	37.1	100	47.9	100	49.5	**100**	47.3	95	54.7
OmniClone	**100**	**36.6**	**100**	**40.3**	**100**	**51.3**	100	**24.4**	100	**20.4**	100	**32.6**	100	**32.9**	95	**32.9**	**100**	**37.7**

(b) Walking and running stratified by speed, jumping by height

Method	**Walk**						**Run**						**Jump**					
	Fast		Medium		Slow		Fast		Medium		Slow		High		Medium		Low	
	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓	SR ↑	MPJPE ↓
OmniClone^{MLP}	20	111.7	35	95.6	85	64.1	90	82.0	80	69.8	70	71.1	75	81.4	85	62.9	75	58.9
OmniClone	**100**	**63.5**	**100**	**48.8**	**100**	**43.7**	**100**	**49.2**	**100**	**42.0**	**90**	**45.5**	75	**50.6**	**100**	**34.5**	**100**	**30.1**

positional discrepancies of 20,cm in reference motions from ***OmniBench***. These perturbations reduce SR by 12.8 percentage points and increase MPJPE by approximately 74.6%. The results demonstrate that scale errors of this magnitude can substantially degrade whole-body tracking, motivating the initial-frame scale calibration used on hardware.

TABLE IX: **Effect of retargeting scale discrepancies.** SR (%) and MPJPE (mm) on reference motions from ***OmniBench***, without perturbation and with a controlled ± 20 cm positional discrepancy.

Metric	0 cm	± 20 cm
SR ↑	97.8	85
MPJPE ↓	39.8	69.5

C. Real-World Experiments

We describe the hardware and deployment setup used in Sec. VI and the protocol of the autonomy experiments in Sec. VI-B, whose complete pipeline for data collection and autonomous execution will be released publicly.

1) System Setup: We deploy ***OmniClone*** on the Unitree G1-Comp platform [38]. A 2-DoF actuated head carries an Intel RealSense D455 depth camera [39] and rotates through $\pm 50^\circ$ in yaw and from -90° to $+22.7^\circ$ in pitch, giving wide visual coverage for manipulation.

Operator motion is captured by a VR headset such as PICO [40] or by an optical MoCap system, retargeted on an external PC, and sent through a wireless router to the robot, where the tracking policy runs on the onboard computer (PC2).

2) Data Utility for Autonomy:

a) Model and tasks: We use GR00T N1.6 [17], a vision-language-action (VLA) foundation model that maps images, language instructions, and robot states to actions, and evaluate it on the standing and squatting pick-and-place tasks. During autonomous execution, ***OmniClone*** is the low-level controller, and communication and command execution run on the same infrastructure as teleoperation. The setup therefore jointly tests (i) the fidelity of demonstrations collected through our teleoperation pipeline, (ii) the reliability of the tracking policy as the low-level controller of the VLA model, and (iii) the robustness of the system infrastructure in executing the commands of the VLA model.

b) Recorded data streams: Our whole-body data collection pipeline is compatible with GR00T N1.6 and records three streams. Visual observations are egocentric RGB images from the head-mounted camera at 30 Hz, temporally interpolated to 50 Hz to synchronize them with the whole-body commands. Proprioceptive states comprise the 29 body-joint positions, the states of the 12 Inspire hand DoFs, the waist orientation, and the 2-DoF head pose. Whole-body control commands are derived from the retargeted motion and mirror the structure of the proprioceptive states for accurate demonstration replay and policy supervision.

c) VLA post-training: We post-train GR00T N1.6 as the high-level planner on the 50 demonstrations collected with ***OmniClone***, following its standard post-training protocol and keeping the vision-language backbone frozen. Training runs for about 20,000 steps on four NVIDIA A100 GPUs with a global batch size of 384 and a learning rate of 1×10^{-4}.

d) Inference protocol: At each query, GR00T N1.6 takes the current image, the language task description, and the proprioceptive state of the robot and predicts a 16-step action chunk with four denoising steps. Forward kinematics converts the joint-space actions of each chunk into body-link positions, which serve as reference commands for ***OmniClone***. This conversion can introduce noise into the references. ***OmniClone*** executes the first eight actions of each chunk before the next query, in a receding-horizon manner. Table IV reports the resulting success rates.